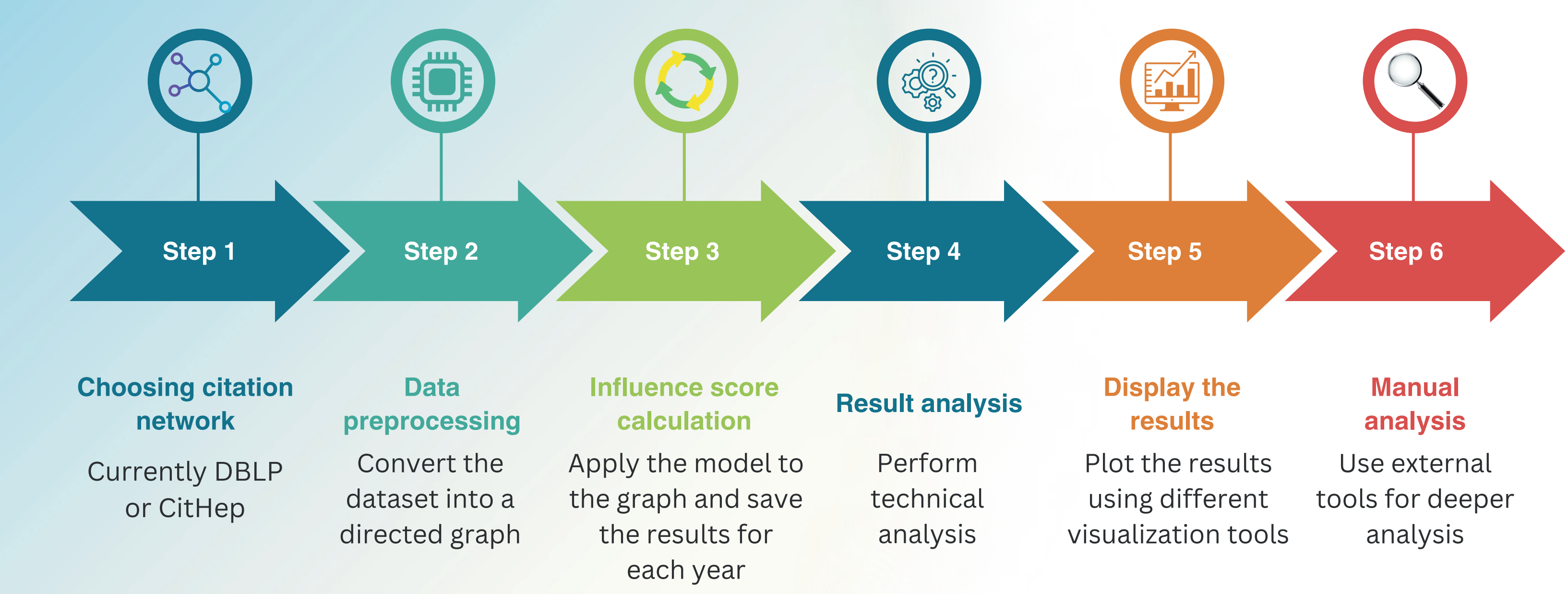
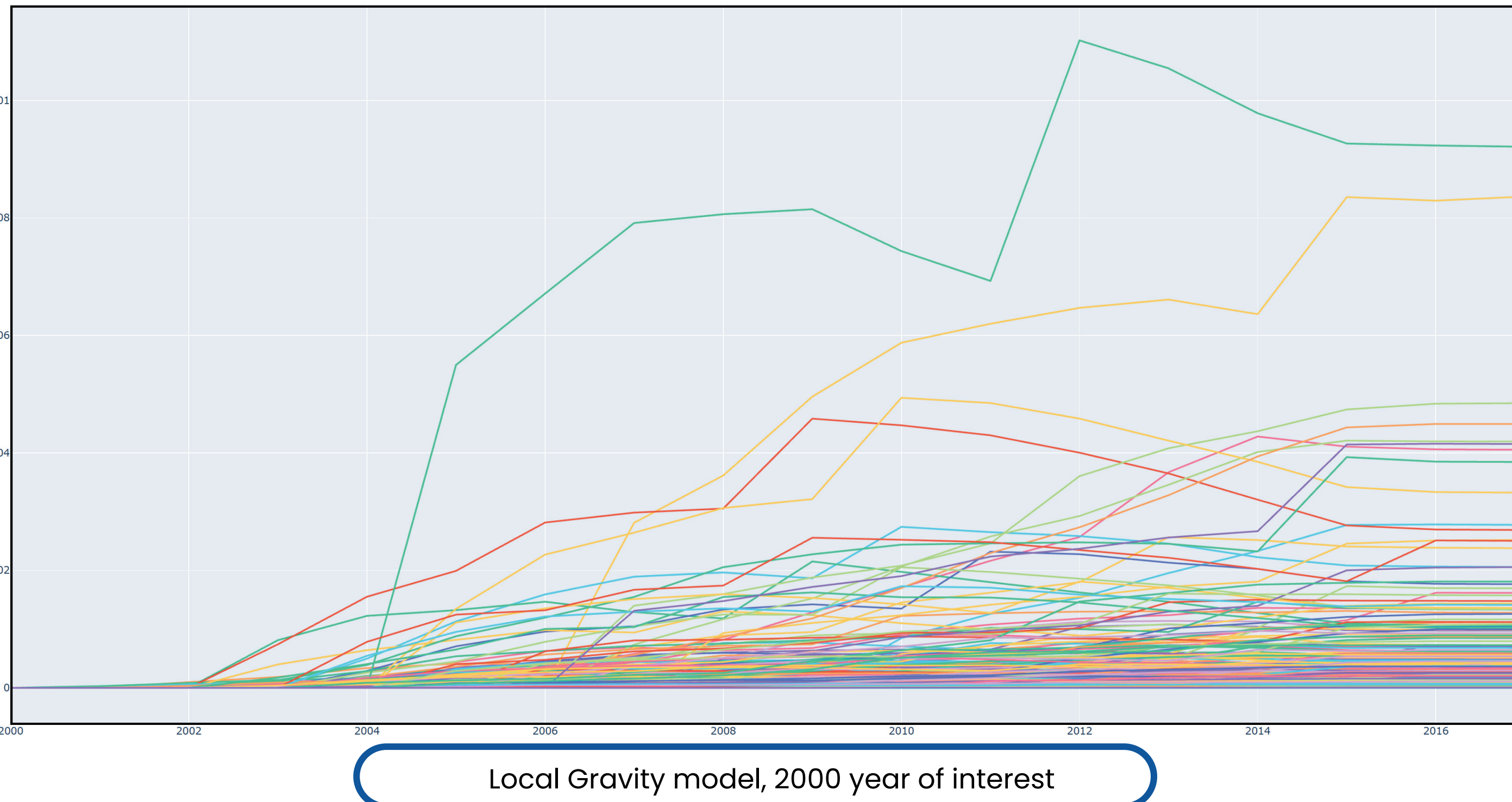
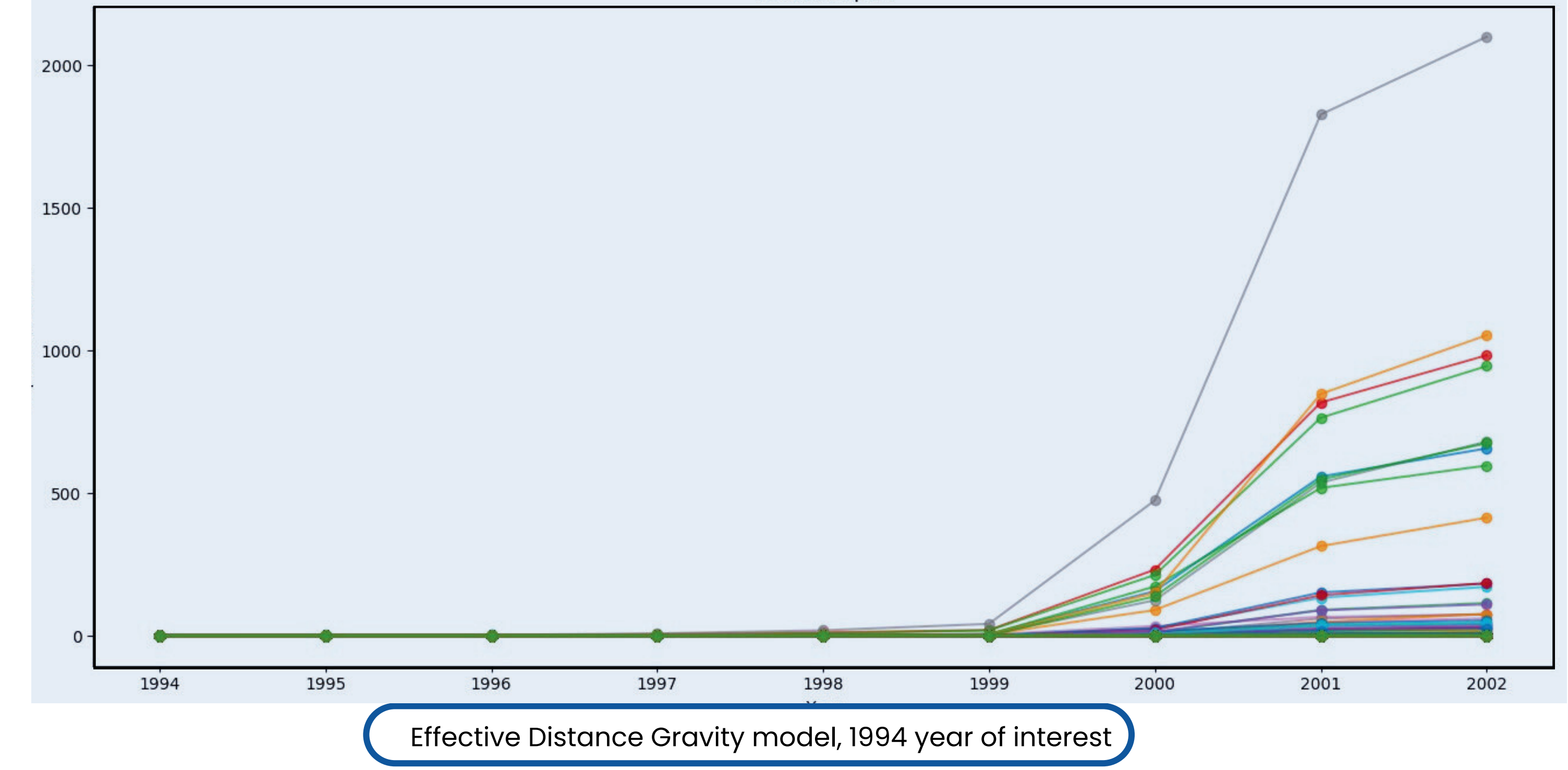
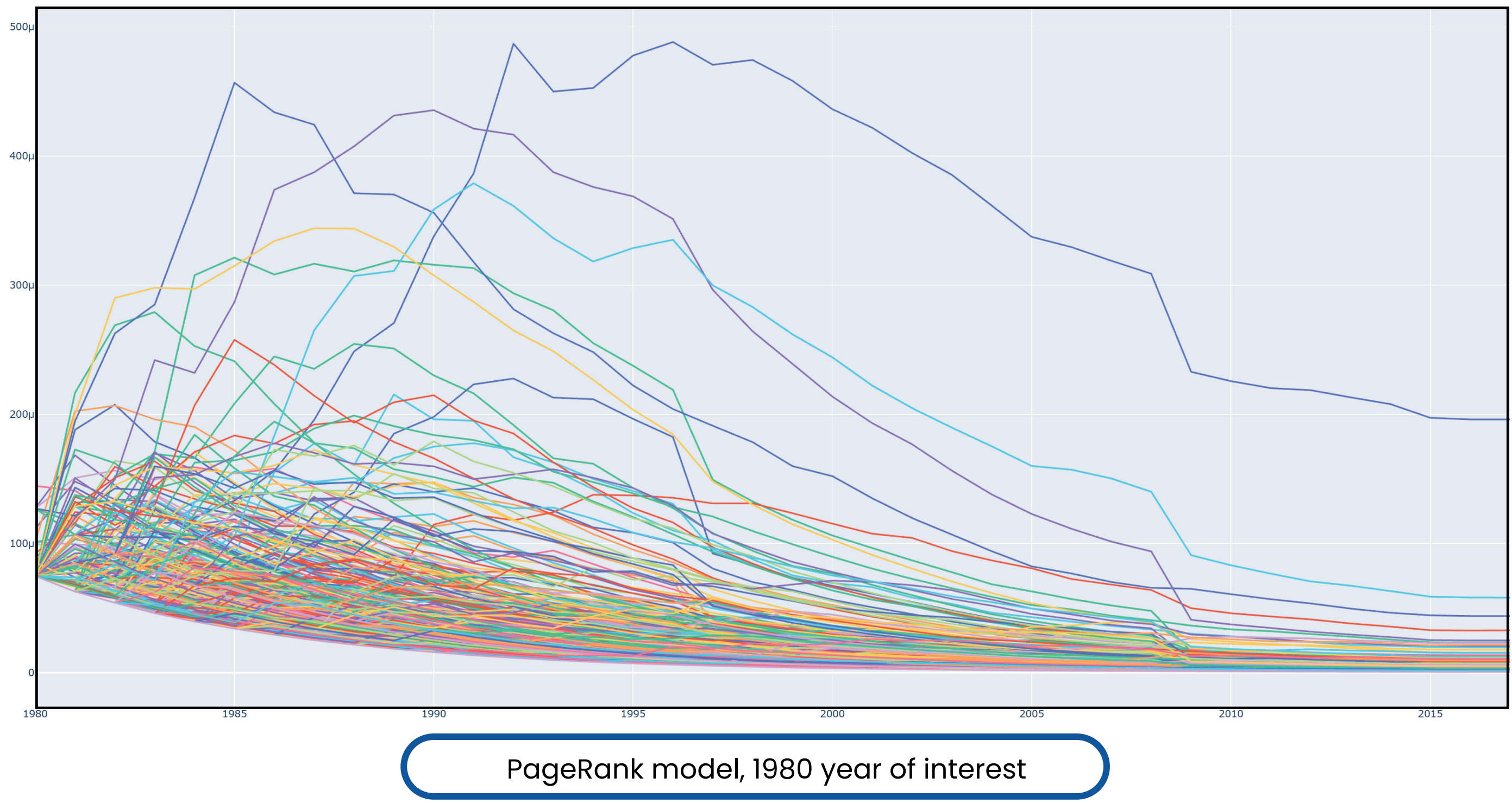
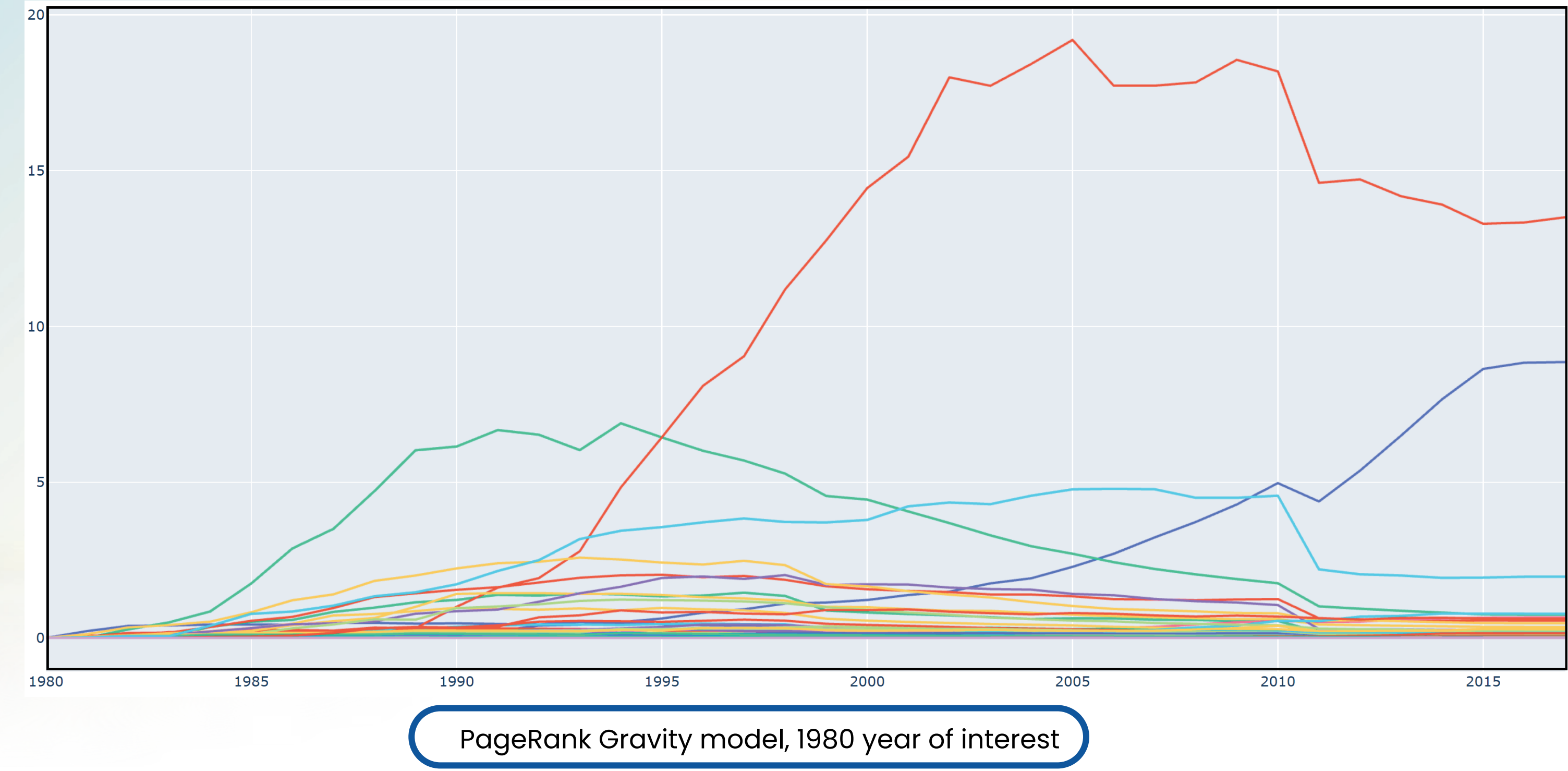


TEMPORAL IDENTIFYING OF INFLUENTIAL NODES IN COMPLEX CITATION NETWORKS



Project goals:

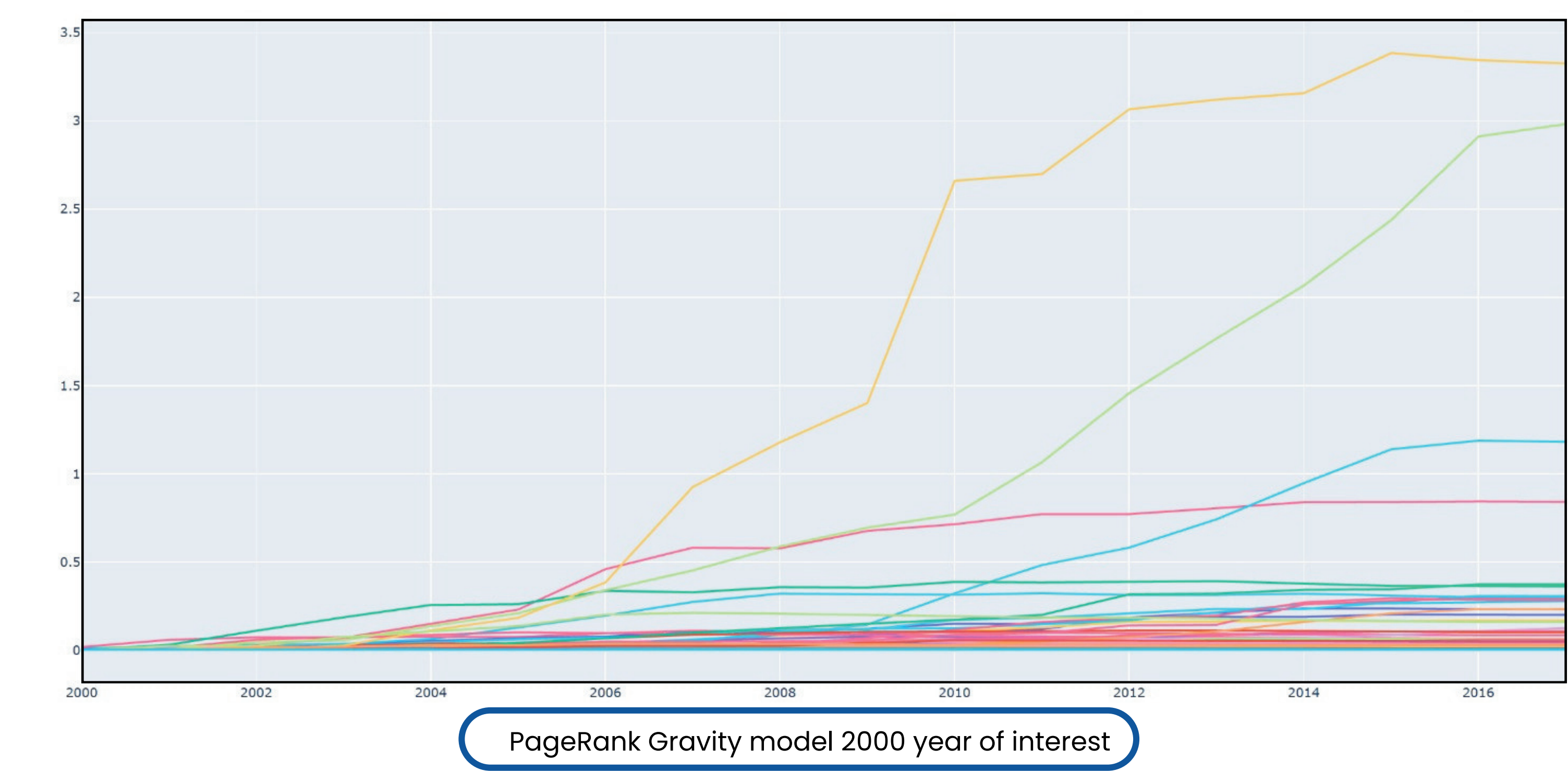
- Using gravity models to analyze papers behavior in citation networks.
- Develop a tool to analyze how the influence of each paper changes over the years.
- Research must contain different Gravity models and datasets.



Project need:

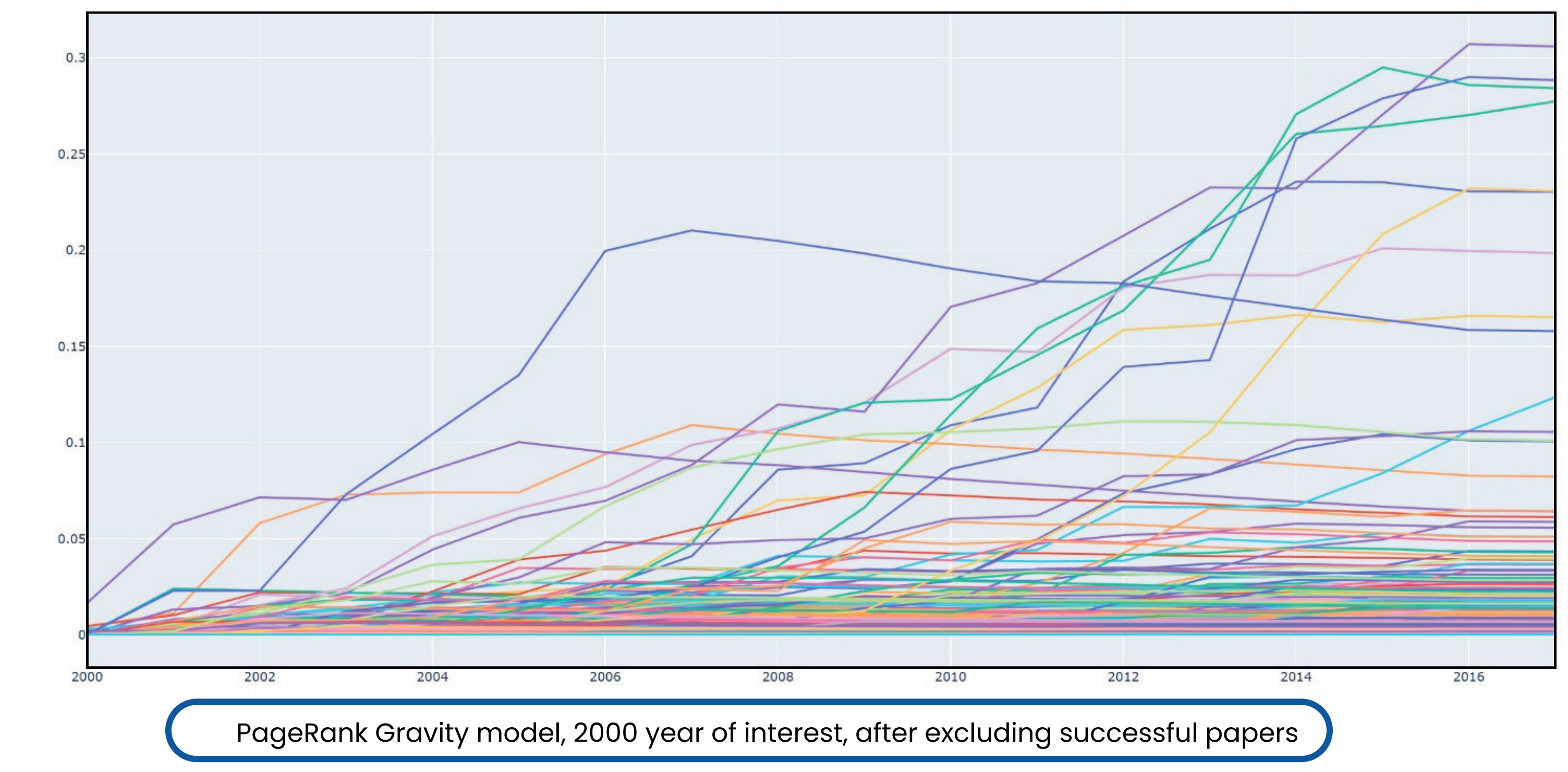
- A tool for evaluating a scientist's eligibility for awards by measuring the impact of their publications on the field.
- Identifying scientific trends, breakthrough ideas, or shifts in research focus to guide future studies.
- Helping the startups to forecast the direction of science and allocate resources effectively (e.g., funding, collaborations, and investments in specific fields).
- Improving academic learning program in higher education by taking in account the most relevant and impactful scientific publications.

1. Execution of the PageRank Gravity model on DBLP dataset



During the research, it was discovered that papers exhibit certain behavior: most papers tend to cluster around the bottom of the plot, while a few successful papers rise sharply. This is a statement about the nature of papers in general. In order to support this claim, the following experiment was conducted:

2. Another execution of the same model after removing the successful papers



3. Additional removal of the successful pages followed by another execution

